9.6.21 AC Three phase single layer lap Winding

AC, 1ϕ, 4 poles, 24 slots single layer lap winding diagram for an induction motor

$$\text{Pole pitch} = \frac{\textit{Number of slots}}{\textit{Number of poles}}$$

$$= \frac{24}{4}$$

$$= 6$$

$$\text{Coil pitch} = 1 + \text{pole pitch}$$

$$= 1 + 6 = 7$$

$$= 1\text{-}7 \text{ (full pitched)}$$

1-6 (short pitched)

In single layer winding, successive coils must be placed by leaving one slot between them. Hence the left coil sides should be inserted in odd slots and the right coil sides should be inserted in even slots.

So, coil pitch = 1 – 8 (or)

= 1- 6

$$\text{Total no. of coils} = \frac{\textit{No. of slots}}{2}$$

(Single layer)

$$= \frac{24}{2}$$

$$= 12$$

$$\text{No. of coils per pole} = \frac{\textit{No. of coils}}{\textit{No. of poles}}$$

(coils / pole)

$$= \frac{12}{4}$$

$$= 3$$

$$\text{No of coils per phase} = \frac{\textit{No. of coils}}{\textit{phases}}$$

(coils / phase)

$$= \frac{12}{3}$$

$$= 4$$

No. of coils per pole per phase

$$= \frac{\textit{No. of coils}}{\textit{poles} \times \textit{phases}}$$

(coils / pole / phase)

$$= \frac{12}{4 \times 3}$$

$$= \frac{12}{12}$$

$$= 1$$

Because of the coils per pole per phase is a whole number, this winding is a balanced winding.

Total no. of coil groups = poles × phases

= 4 × 3 = 12

No. of coils per coil groups

$$= \frac{\textit{No. of coils}}{\textit{No of coil groups}}$$

(coils / coil group)

$$= \frac{12}{12}$$

$$= 1$$

Pole1			Pole2			Pole3			Pole4		
R	Y	B	R	Y	B	R	Y	B	R	Y	B
1	1	1	1	1	1	1	1	1	1	1	1

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